

Markov Chains

Exercises: Chapter 2 and 3.

Ex1. (Study the Green function) Consider a $MC(I, P)$. Let

$i, j \in I$ be two states show that

a) If $i \nrightarrow j$ then $G(i, j) = 0$.

b) If $i \rightarrow j$ and

- j is transient, then $G(i, j) < \infty$.

- j is recurrent, then $G(i, j) = \infty$.

c) The function G satisfies the equation

$$G = PG + I \Leftrightarrow \forall (i, j) \in I^2, G(i, j) = \delta_{i=j} + \sum_{k \in I} p_{ik} G(k, j).$$

Ex2. Consider $MC(I, P)$ with $I = \{1, 2, 3\}$ and

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1/2 & 1/2 & 0 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$$

a) Draw the graph of the MC and classify the states

(recurrent / transient).

b) Calculate the Green function for all point $(i, j) \in I^2$.

Deduce that the prob of never coming back to 3 given that we started from 3.

Ex3. Consider a player whose probability of giving a correct answer is p whatever the question asked and the probability of giving wrong answer is $1-p$, $0 < p < 1$. We assume that the answers are independent of each other, and if he give a correct answer, he wins 1, he loses everything if his answer is incorrect. The starting wealth is $X_0 = 0$ and for all $n \geq 1$, we note that X_n is his wealth at time n .

- Show that $(X_n)_{n \geq 0}$ is a Markov chain. What is the state space? Give its graph and transition matrix.
- Show that the MC is irreducible.
- Let $T_0 = \inf\{n > 0 \mid X_n = 0\}$ the time to reach 0. $\forall n \in \mathbb{N}^*$, calculate $P_0(T_0 = n)$.
- Deduce that the chain is recurrent.

Ex4. We consider the MC $(X_n)_{n \geq 0}$ with state space $E = \{1, 2\}$ and transition matrix $P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$, $0 < p, q < 1$.

- Classify the states of the chain.
- Determine the stationary probability
- Deduce the expectation of return times to state 1 knowing that it started from 1.

Ex 5. Consider again the MC in exercise 3.

- Find a stationary prob for this MC.
- Deduce that the chain is positive recurrent and give the expectation of return times to 0 given that we started from 0.

Ex 6. Consider the following matrix on the state space

$$E = \{1, 2, 3, 4, 5, 6\}.$$

$$P = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{1}{2} & a & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & 0 & \frac{3}{4} \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}$$

a) Find the value of a

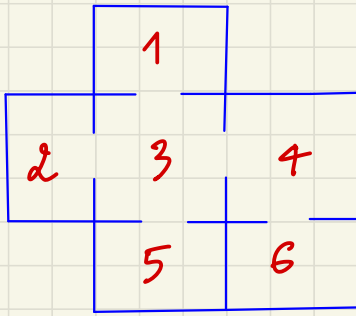
so that P is a transition matrix.

b) Draw the graph.

c) Give the communication classes. Determine their period and if they are recurrent or transient.

d) We note $T_4 := \inf\{n \geq 0, X_n = 4\}$. Find $P_5(T_4 < +\infty)$.

Ex 7. A mouse runs through the maze shown below:



At each step, when it leaves the room it is in by choosing at random one of the doors out of the room.

- Give the transition matrix P for this MC.
- Show that the MC is irreducible but not aperiodic.
- Find the stationary distribution.
- Now suppose that a piece of cheese is placed on a deadly trap in Room 5. The mouse starts in Room 1. Find the expected value of steps before reaching Room 5 for the first time, starting in Room 1.
- Find the expected time to return to Room 1.